

## The problem

Epigenomic measurements often suffer from noise, batch effects and irreproducibility. Existing imputation methods operate on idealized processed signal, and assume that this processing removes all batch effects. In the ENCODE Imputation Challenge, due to subtle differences between the train and test sets, a simple baseline outperformed most of the entrants [1].

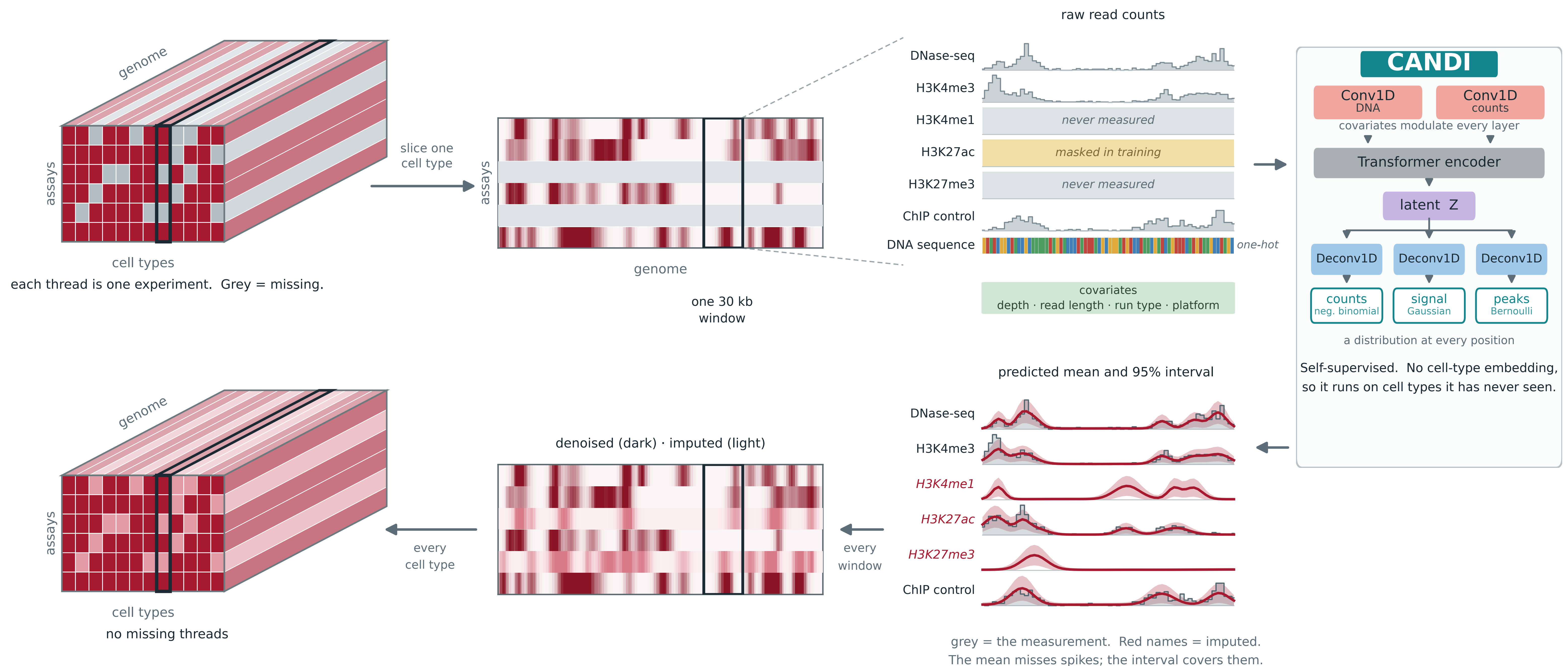
## CANDI

**Confidence-Aware Neural Denoising Imputer** [2] is a self-supervised deep learning method for epigenome imputation and denoising. It takes raw read counts, DNA sequence and experiment-specific covariates as input, and outputs probability distributions over raw counts, signal values and peak calls. By conditioning on experimental metadata rather than learned cell-type embeddings, it enables zero-shot generalization to unseen cell types.

## Key findings

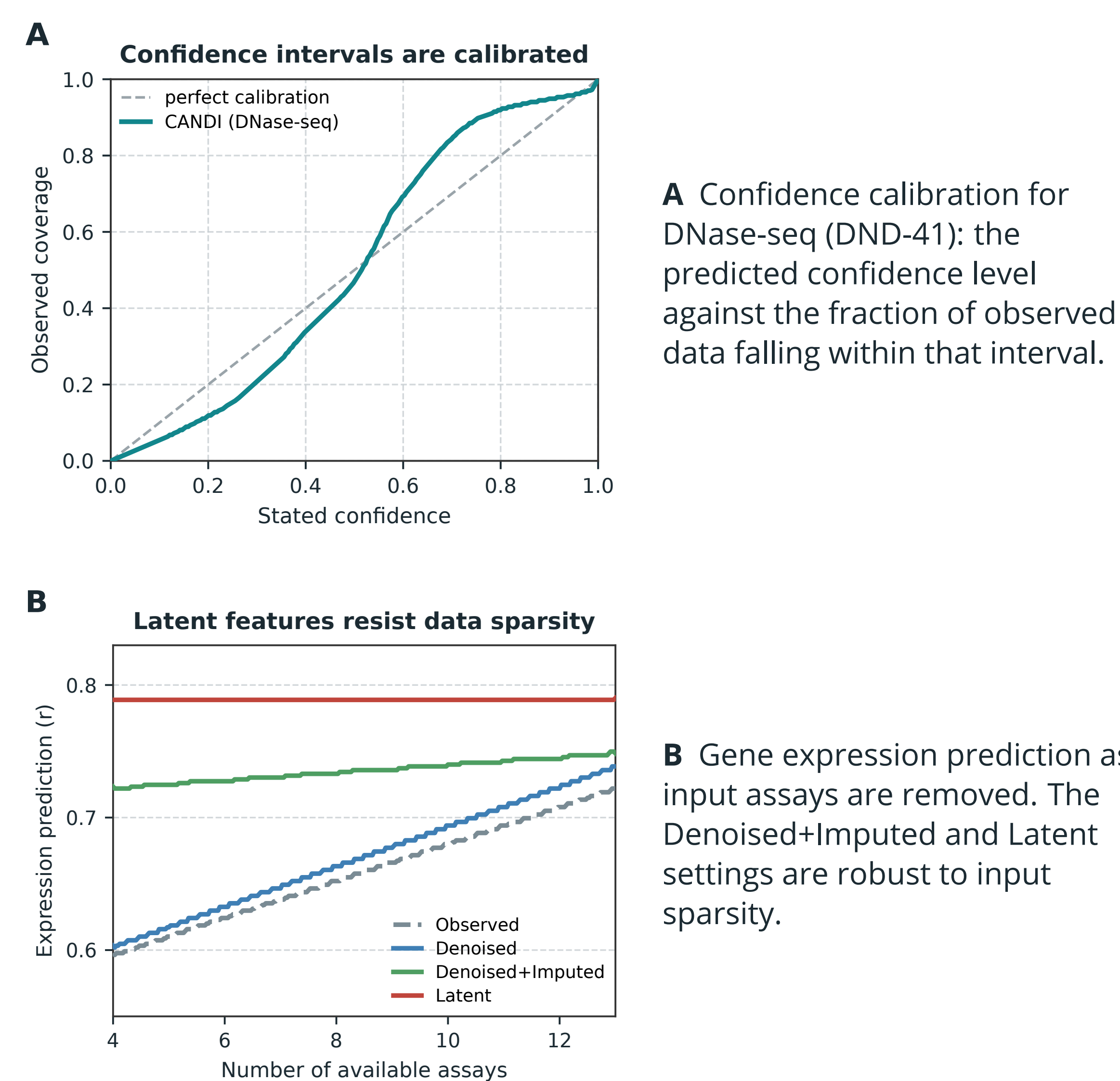
- CANDI ranks **1st of 26** on the ENCODE Imputation Challenge blind test.
- CANDI provides calibrated uncertainty estimates.
- CANDI's imputed signals and latent space predict gene expression.

## CANDI schematic workflow



**Top:** for a given 30 kb locus, CANDI takes as input the observed read counts with the ChIP-seq control, four experimental covariates per assay and the DNA sequence; grey assays were never measured, and yellow ones are masked in training. **Bottom:** all outputs are probability distributions, here a mean and a 95% interval for every assay; window by window, they fill every missing thread of the tensor.

## Downstream utility of CANDI

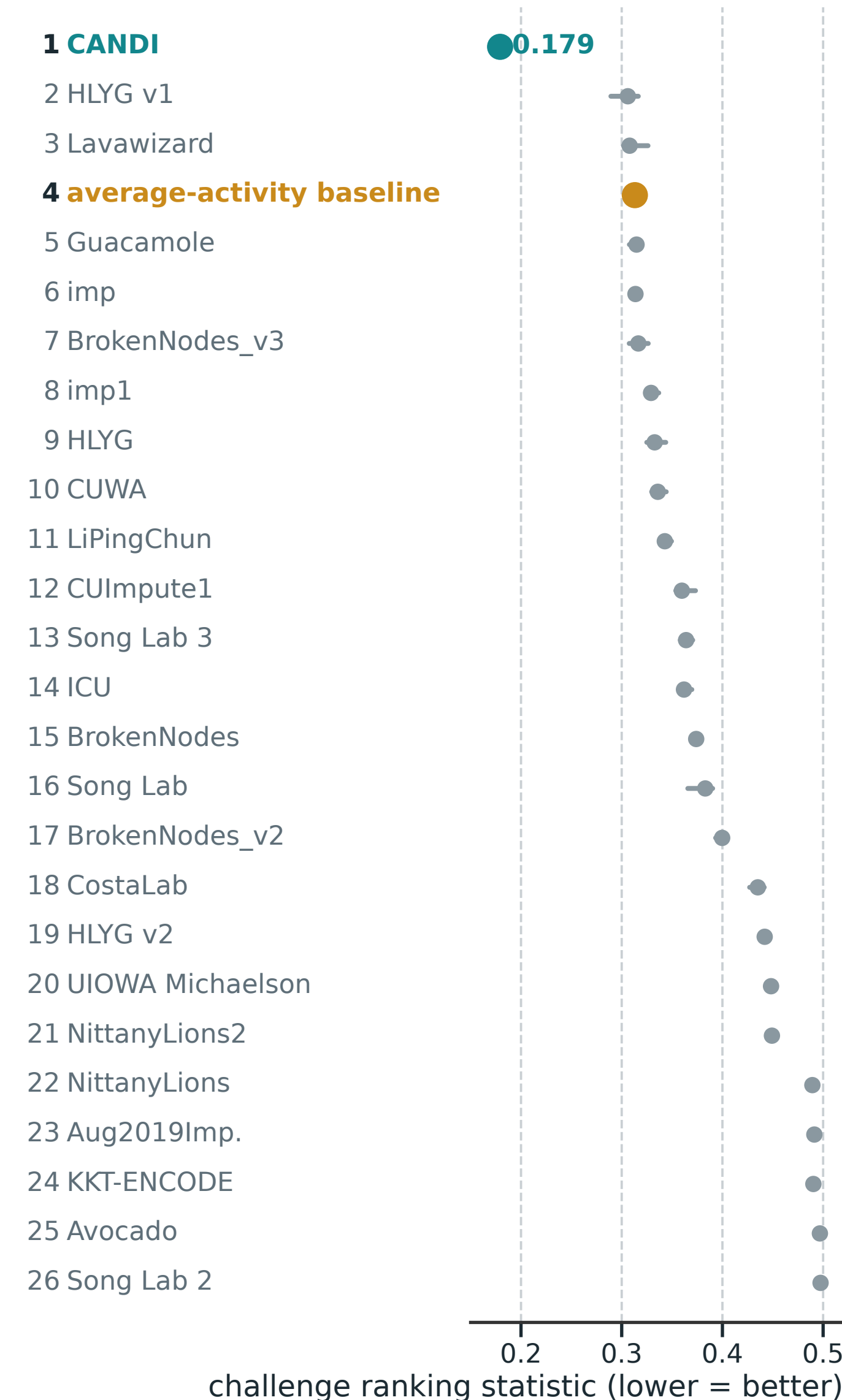


[1] Schreiber J, Boix C, *et al.* The ENCODE Imputation Challenge: a critical assessment of methods for cross-cell type imputation of epigenomic profiles. *Genome Biol* 24:79 (2023).

[2] Foroozandeh M, Diab AR, Libbrecht M. CANDI: self-supervised, confidence-aware denoising imputation of genomic data. *bioRxiv* 2025.01.23.634626 (2025).

Panels C–D: blind-test scoring from the Libbrecht lab's EIC reproduction (experiments 017, 025). *Baseline* is the challenge's average-activity track. The output correction was chosen after the challenge results were public; none of its numbers is fitted on blind-test data.

## Fixing the signal-level shift in the ENCODE Imputation Challenge blind test



**C The challenge's leaderboard with CANDI added.** CANDI is first at 0.179 (bootstrap range 0.177–0.184); the next entry scores 0.306. Bars span the ten bootstrap groups.

**The shift.** We followed the train-test split of the EIC: 51 blind-test experiments (12 cell types, 8 assays). There are subtle differences between the train and test sets: every blind-test experiment is paired-end, against 15% of the training experiments, and the blind-test tracks sit at an assay-specific signal level (3.3× for H3K4me3, 1.7× for H3K27ac, 0.9–1.0× for broad marks). CANDI recovers the rank-ordering and structural patterns of the signal but not its absolute magnitude, and its raw output ranks 14th.

**The fix.** For each assay, CANDI's output passes through one layer,

$$\max(\beta_0 + \beta_1 \cdot \text{CANDI} + \beta_2 \cdot \text{baseline}, 0),$$

whose coefficients are fitted by least squares on the 133 training experiments only. With it, CANDI ranks 1st. Our scorer reproduces every published entry to a median relative difference of  $9 \times 10^{-10}$ .

